US012416765B2

(12) United States Patent
Nakamura et al.

(10) Patent No.: US 12,416,765 B2
(45) Date of Patent: Sep. 16, 2025

(54) **FERRULE, OPTICAL CONNECTOR, AND OPTICAL CONNECTOR MODULE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 314 days.

(21) Appl. No.: **18/182,279**

(22) Filed: **Mar. 10, 2023**

(65) **Prior Publication Data**

US 2023/0296849 A1 Sep. 21, 2023

(30) **Foreign Application Priority Data**

Mar. 16, 2022 (JP) 2022-041462

(51) **Int. Cl.**
G02B 6/38 (2006.01)
G02B 6/36 (2006.01)

(52) **U.S. Cl.**
CPC ***G02B 6/381*** (2013.01); ***G02B 6/3636*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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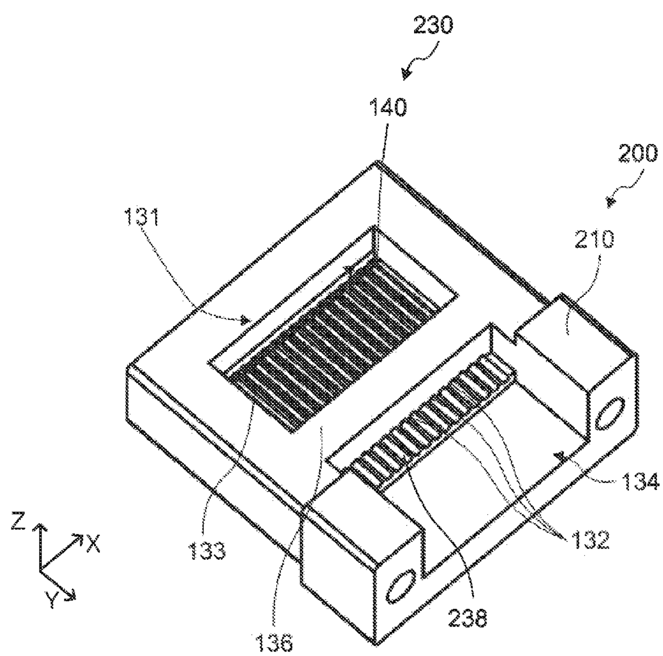
(57) **ABSTRACT**

A ferrule includes a holding part, a first surface and a second surface. The holding part includes a first holding recess including the first surface and a third surface facing the first surface, a first wall, at least one through hole that is open at the third surface and a fourth surface, the fourth surface being located on a side on which the plurality of optical transmission members is inserted, and a plurality of first grooves disposed at the first holding recess along an extending direction of the plurality of optical transmission members, the plurality of optical transmission members inserted to the at least one through hole is respectively disposed at the plurality of first grooves.

9 Claims, 9 Drawing Sheets



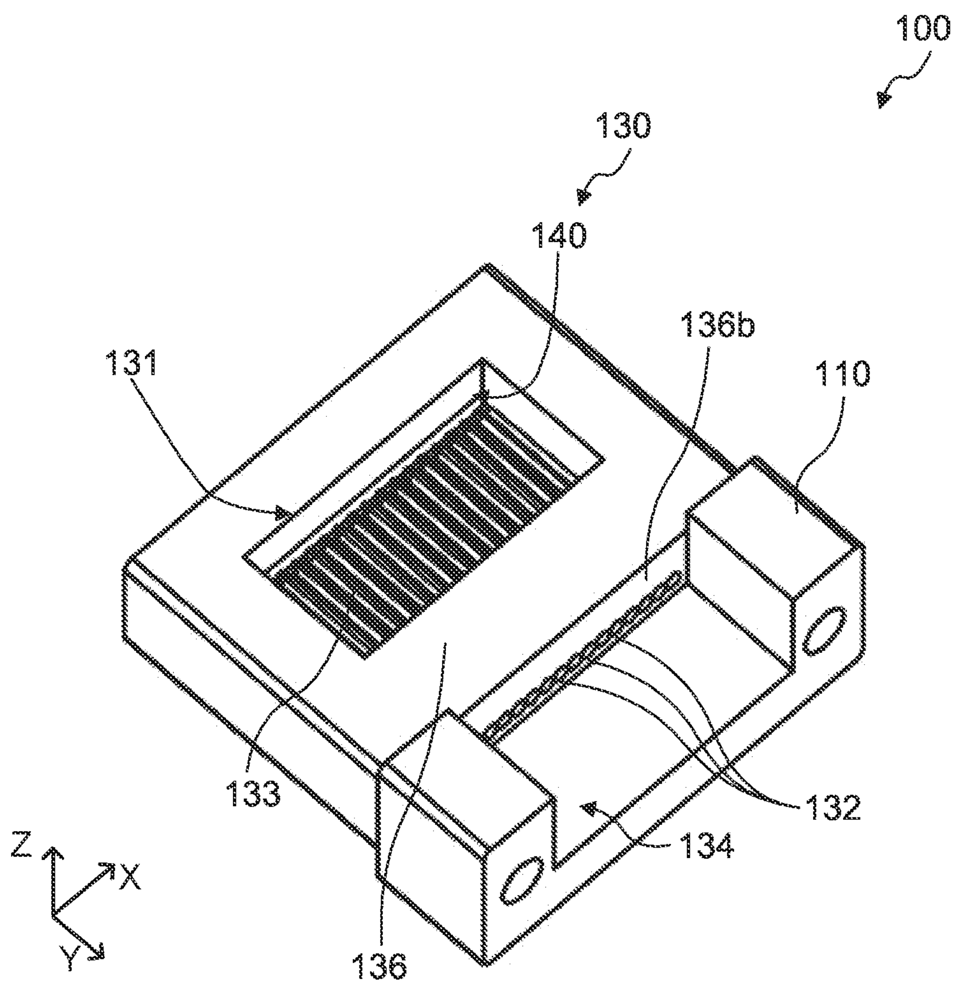
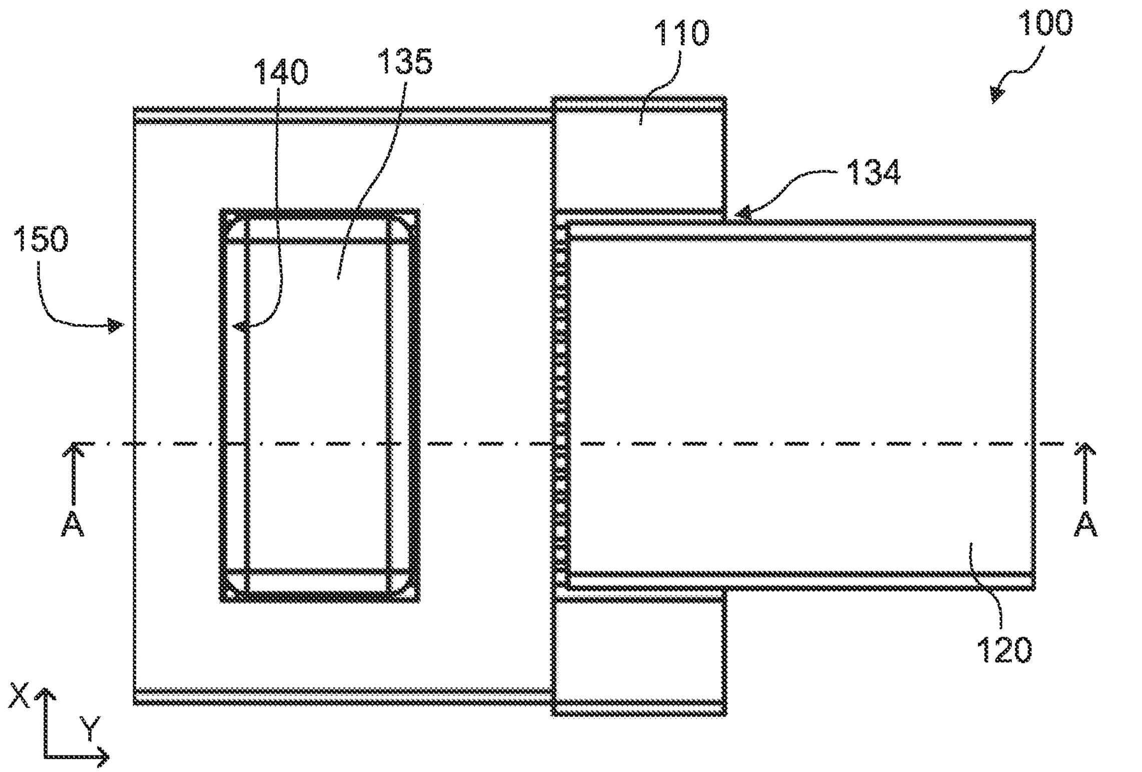
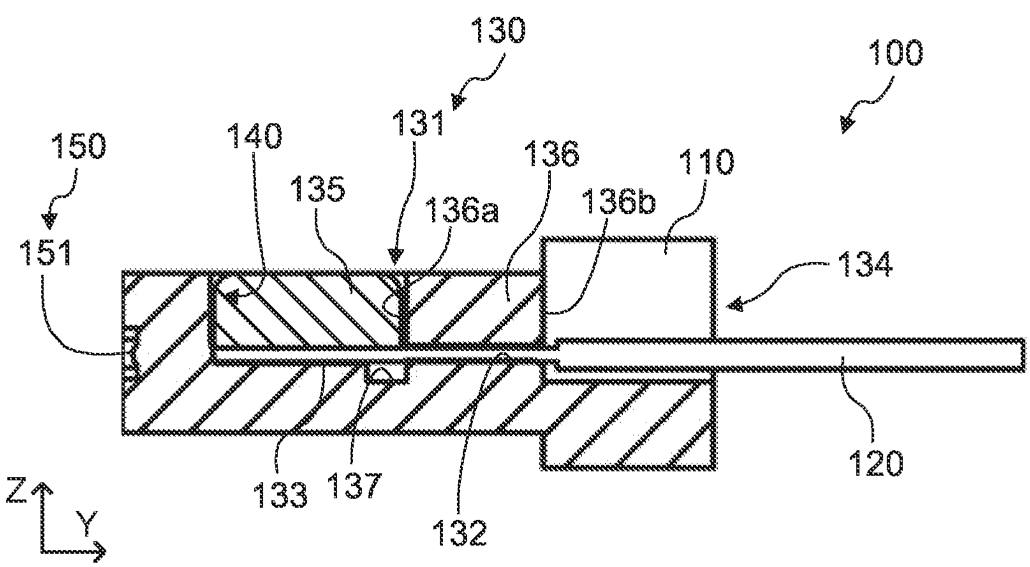


FIG. 1



100
110
140
135
134
150
A
A
120
X
Y

FIG. 2A



130
100
131
150
140
136
110
135
136a
136b
151
134
120
133
137
132
Z
Y

FIG. 2B

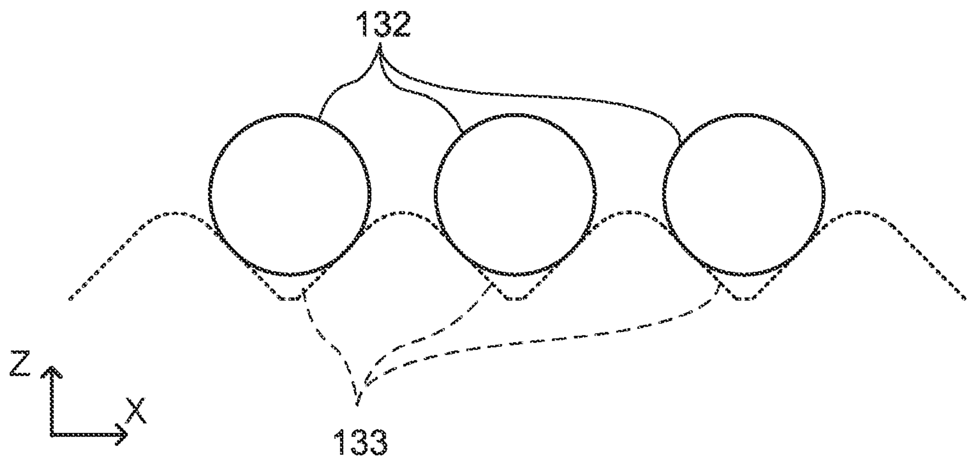


FIG. 3A

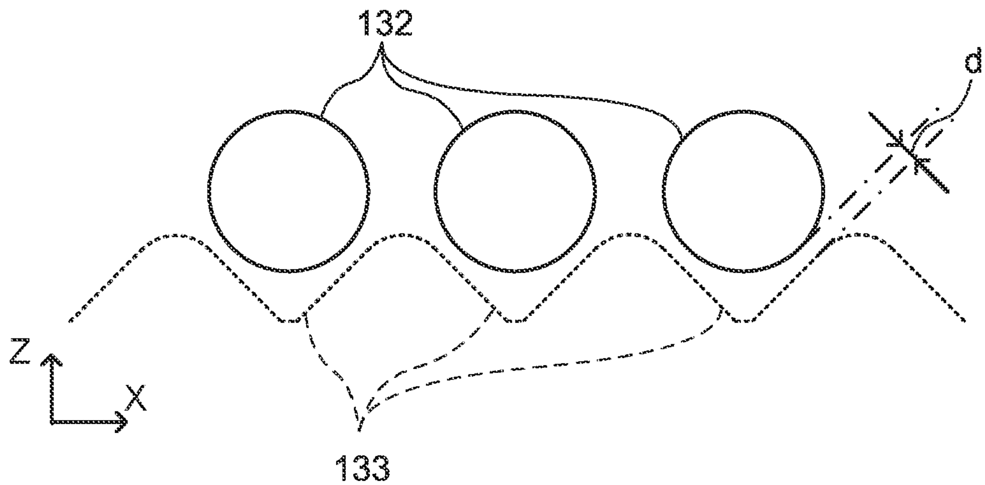


FIG. 3B

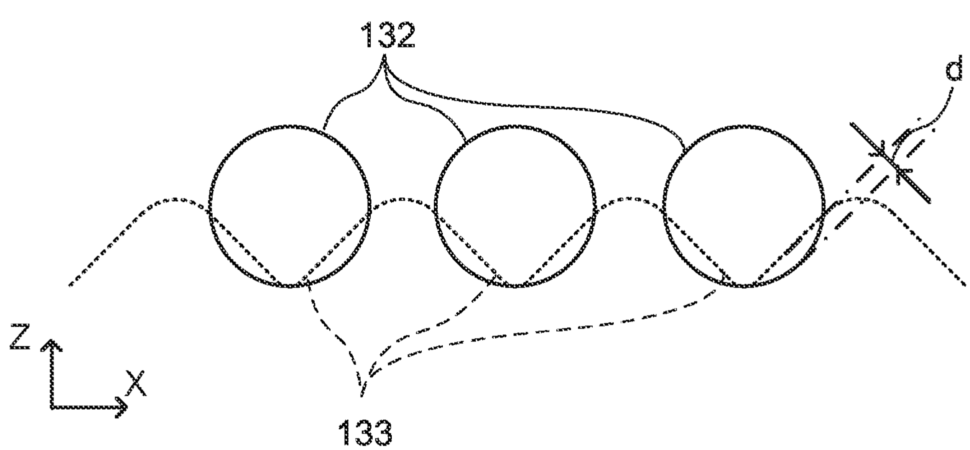


FIG. 3C

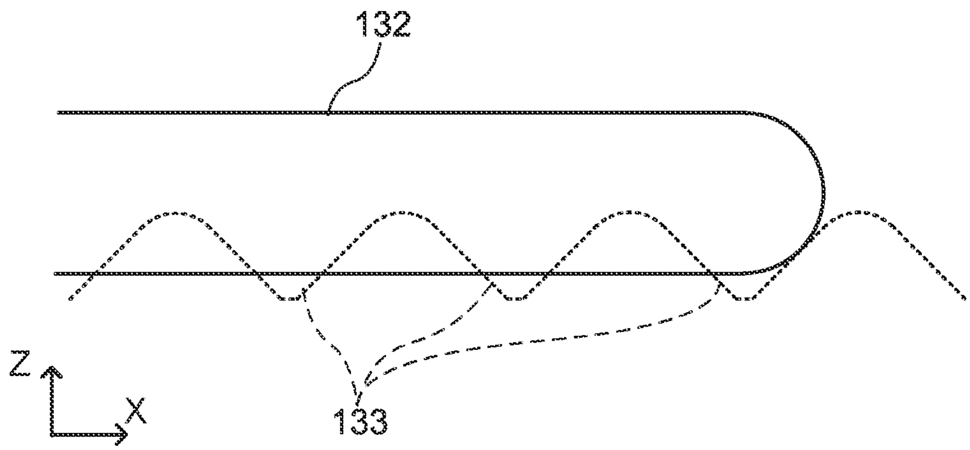


FIG. 4A

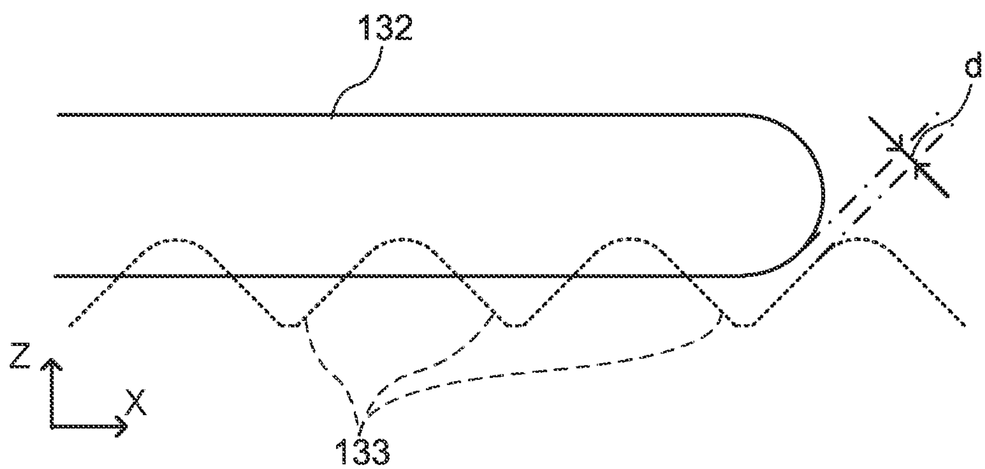


FIG. 4B

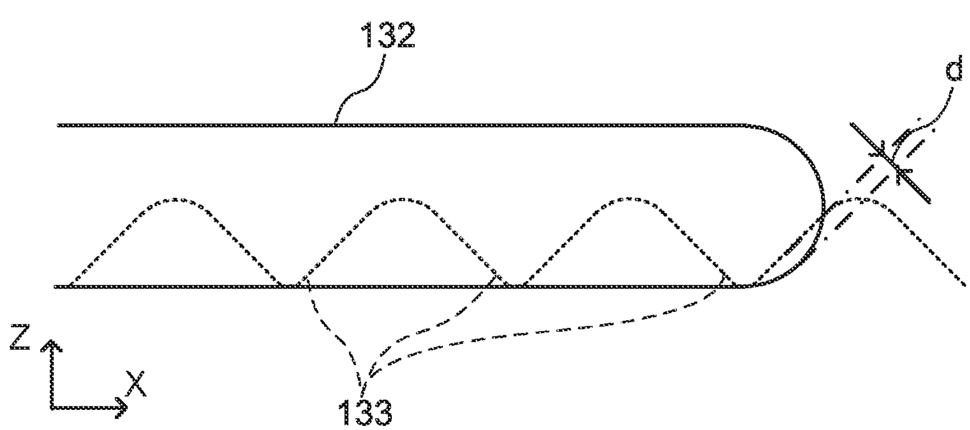


FIG. 4C

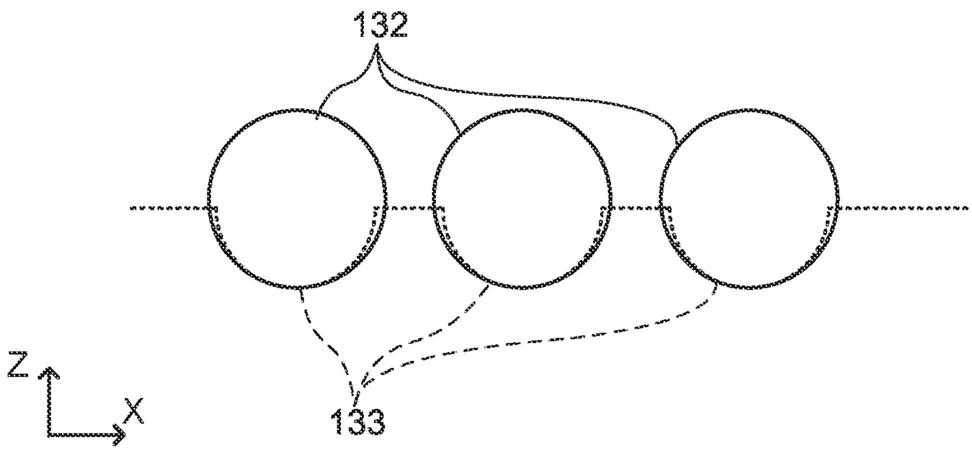


FIG. 5A

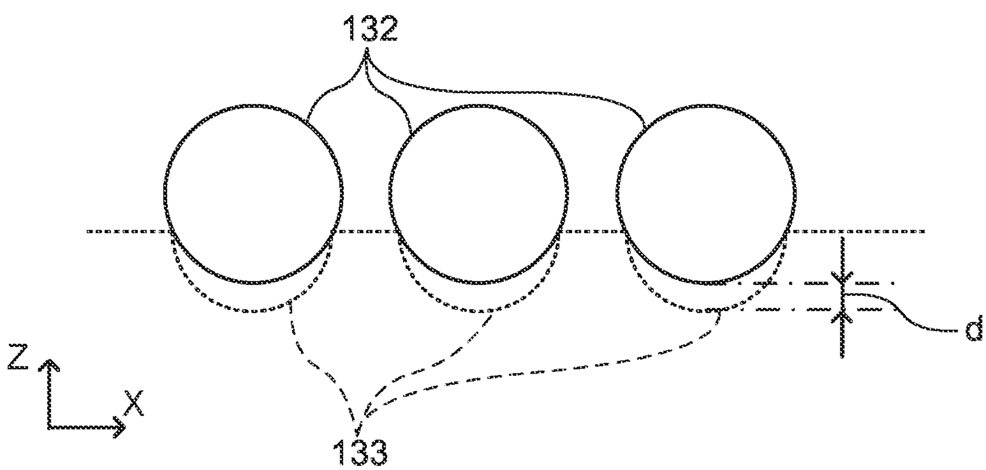


FIG. 5B

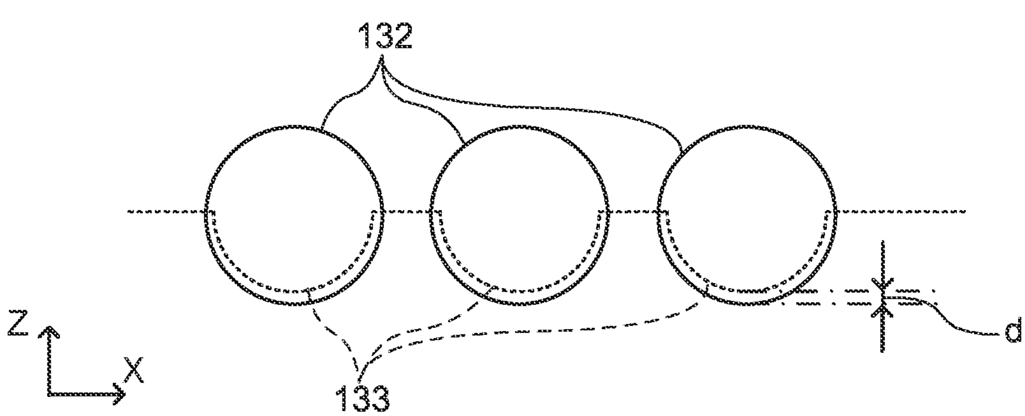


FIG. 5C

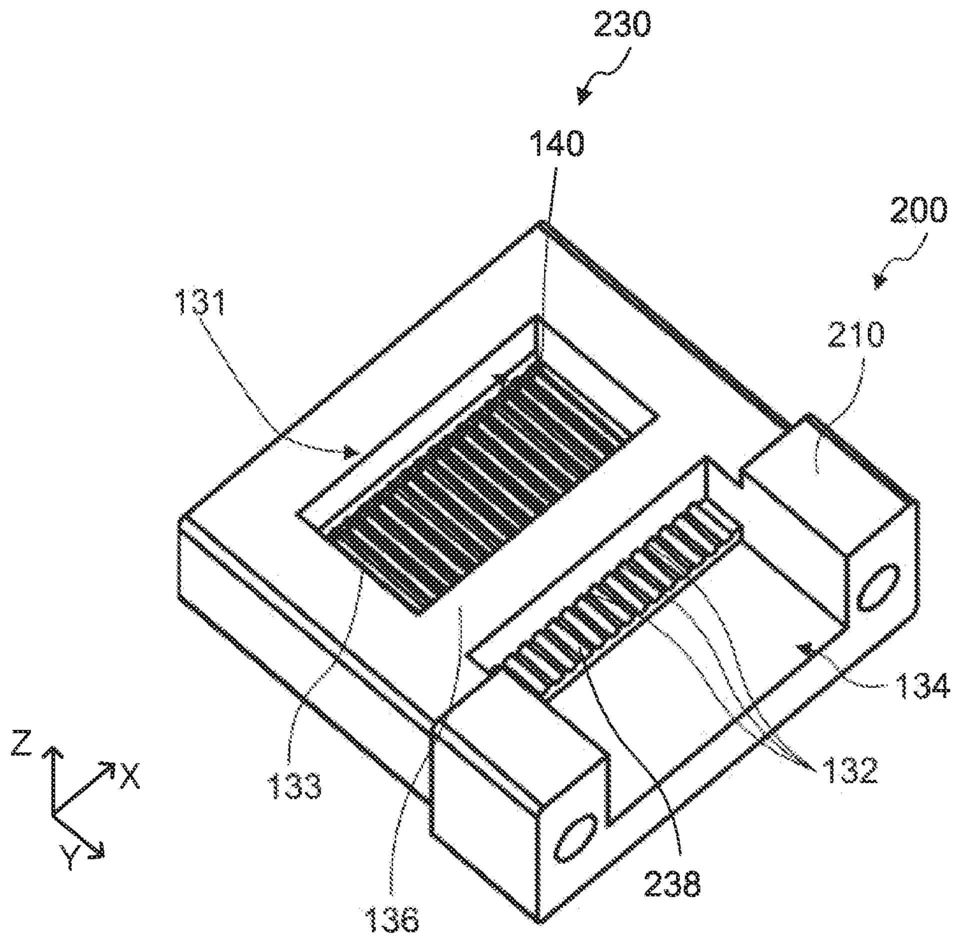


FIG. 6

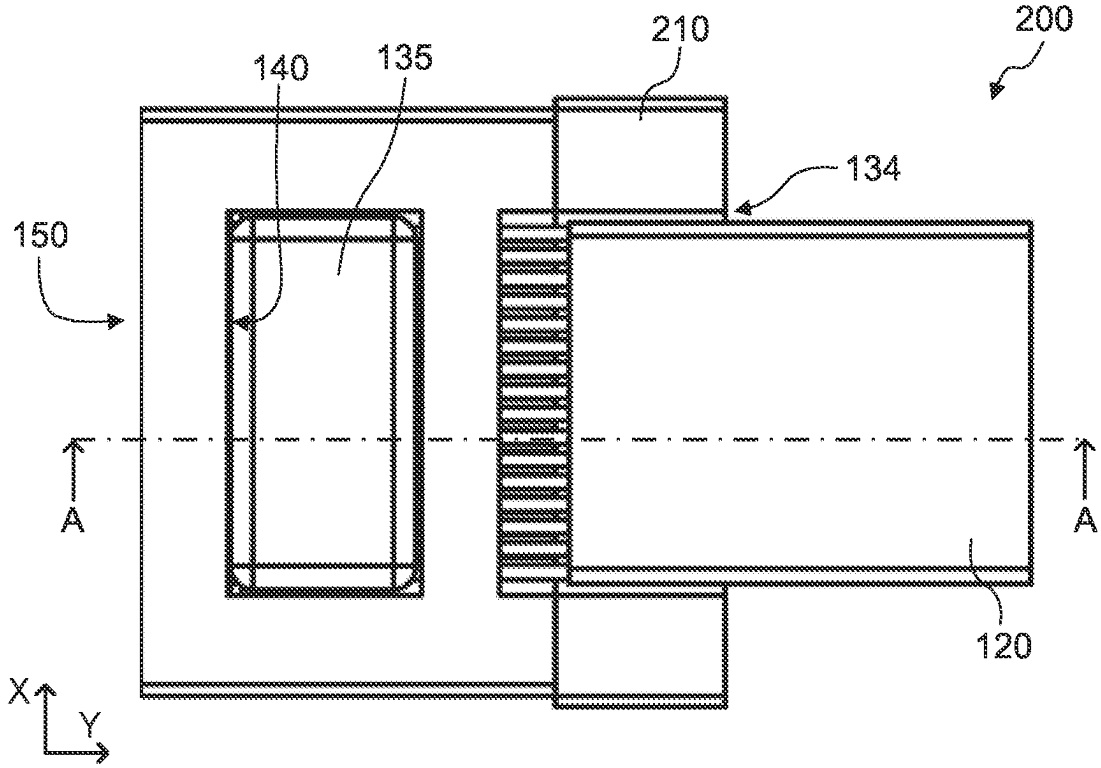

200
210
140
135
134
150
A
A
120
X
Y

FIG. 7A

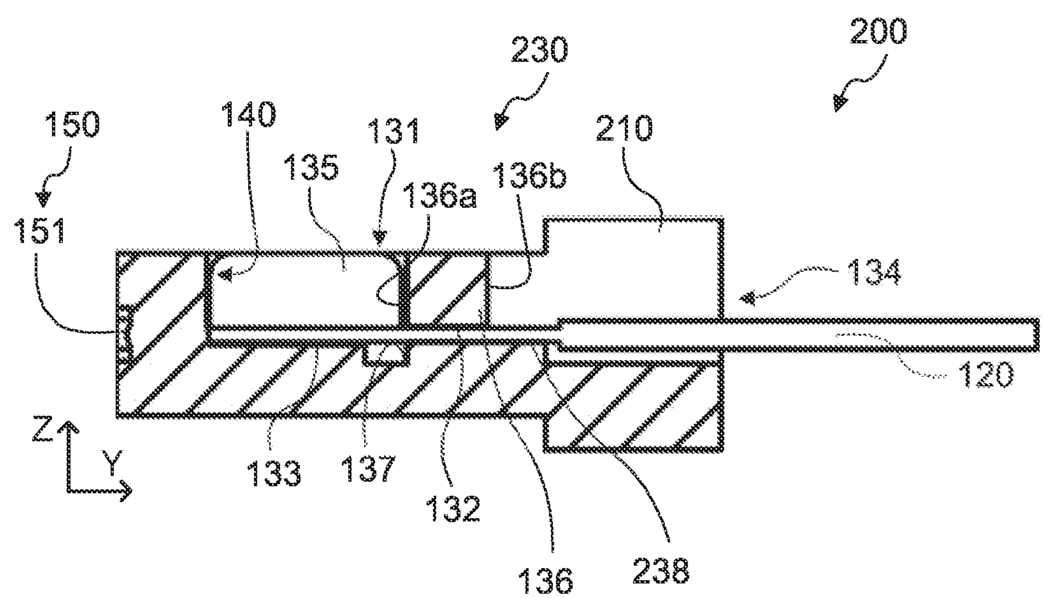

200
230
150
140
131
210
135
136a
136b
151
134
120
Z
Y
133
137
132
136
238

FIG. 7B

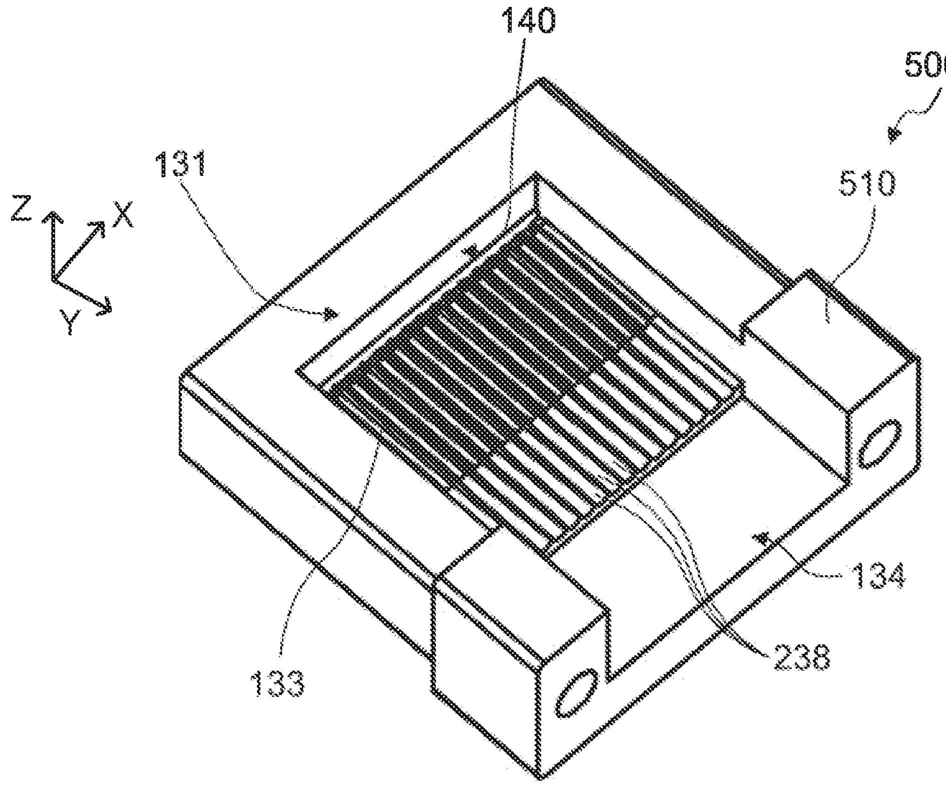
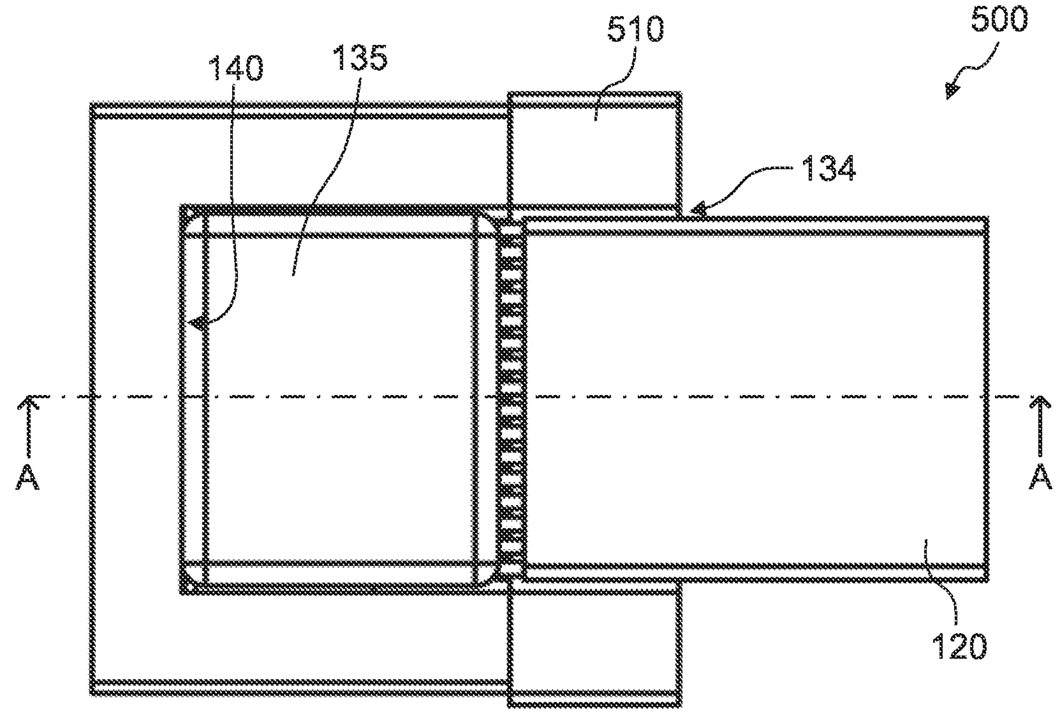
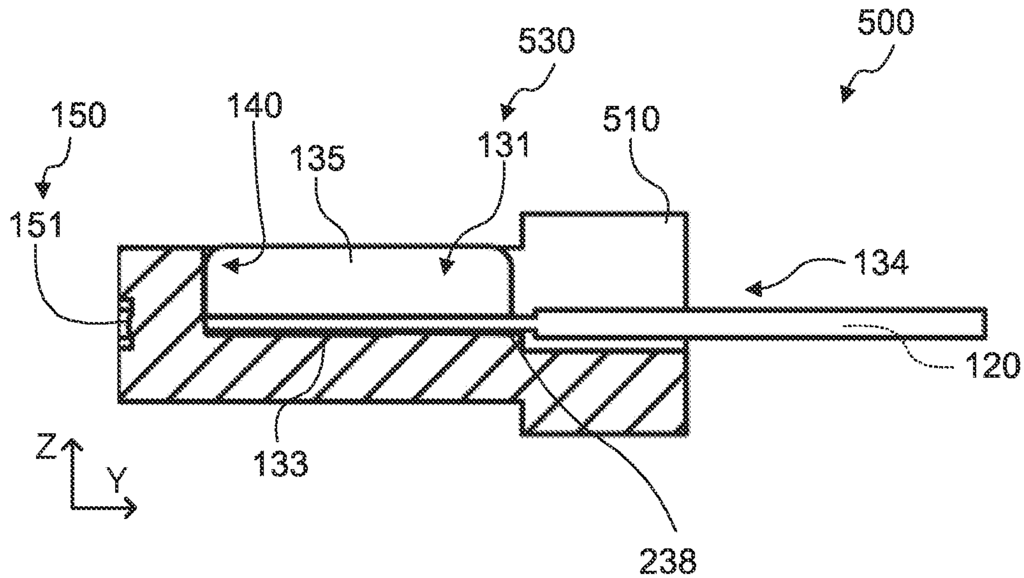


FIG. 8



500
510
134
140
135
A
A
120

FIG. 9A



500
530
150
140
135
131
510
151
134
120
133
238
Z
Y

FIG. 9B

FERRULE, OPTICAL CONNECTOR, AND OPTICAL CONNECTOR MODULE

This application is entitled to the benefit of Japanese Patent Application No. 2022-041462, filed on Mar. 16, 2022, the disclosure of which including the specification, drawings and abstract is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a ferrule for holding an optical transmission member, an optical connector and an optical connector module.

BACKGROUND ART

In the related art, optical transmission members such as optical fibers and optical waveguides are used for optical communications. In the state where the front end is held by a ferrule, an optical transmission member is optically connected to another optical component and another optical transmission member (see, for example, PTL 1).

PTL 1 discloses a base (ferrule) including a light coupling part including a plurality of optical surfaces and serving as a light path between a coupling element and an optical fiber, a fiber holding part, and a fiber supporting part. The fiber holding part includes a V-groove and a fiber holding cover, and holds one end portion of the optical fiber. The fiber supporting part includes a recess and a cover, and supports the optical fiber such that the optical fiber is not bent.

The base disclosed in PTL 1 fixes the optical fiber by pushing the optical fiber to the V-groove from the upper side with the fiber holding cover, and then pressing the optical fiber to the recess from the upper side with the cover.

CITATION LIST

Patent Literature

PTL 1
US Patent Application Publication No. 2019/0079253

SUMMARY OF INVENTION

Technical Problem

However, in the base (ferrule) disclosed in PTL 1 and the like, the clearance between the V-groove and the fiber is small, and consequently it is difficult to correctly dispose the fiber to the V-groove.

An object of the present invention is to provide a ferrule that can accurately position the front end of an optical transmission member, and is easy to assemble. In addition, another object of the present invention is to provide an optical connector and an optical connector module including the ferrule.

Solution to Problem

A ferrule according to an embodiment of the present invention is configured to hold a plurality of optical transmission members, the ferrule including: a holding part configured to hold one end portion of the plurality of optical transmission members; a first surface configured to allow, to enter the ferrule, light emitted from the plurality of optical transmission members held by the holding part, or configured to emit, toward the plurality of optical transmission members, light advanced inside the ferrule; and a second surface configured to emit, to outside of the ferrule, the light advanced inside the ferrule, or configured to allow, to enter the ferrule, light from the outside. The holding part includes: a first holding recess including the first surface and a third surface facing the first surface, at least one through hole that is open at the third surface and a fourth surface, the fourth surface being located on a side on which the plurality of optical transmission members is inserted, and a plurality of first grooves disposed at the first holding recess along an extending direction of the plurality of optical transmission members, the plurality of optical transmission members inserted to the at least one through hole is respectively disposed at the plurality of first grooves.

An optical connector according to an embodiment of the present invention includes the ferrule; and a plurality of optical transmission members held by the ferrule.

An optical connector module according to an embodiment of the present invention includes an optical connector of an embodiment of the present invention.

Advantageous Effects of Invention

According to the present invention, a ferrule that can accurately set the position of the end of the optical transmission member and is easy to assemble can be provided.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view of an optical connector according to Embodiment 1 of the present invention;

FIGS. **2**A and **2**B illustrate a configuration of the optical connector according to Embodiment 1 of the present invention;

FIGS. **3**A to **3**C illustrate a positional relationship between a first groove and a through hole;

FIGS. **4**A to **4**C illustrate a positional relationship between the first groove and the through hole;

FIGS. **5**A to **5**C illustrate a positional relationship between the first groove and the through hole;

FIG. **6** is a perspective view of an optical connector according to Embodiment 2 of the present invention;

FIGS. **7**A and **7**B illustrate a configuration of an optical connector according to Embodiment 2 of the present invention;

FIG. **8** is a perspective view of an optical connector according to a reference example; and

FIGS. **9**A and **9**B illustrate a configuration of an optical connector according to a reference example.

DESCRIPTION OF EMBODIMENTS

A ferrule, an optical connector and an optical connector module according to an embodiment of the present invention are elaborated below with reference to the accompanying drawings.

Embodiment 1

Configuration of Optical Connector

FIG. **1** is a perspective view of optical connector **100** according to Embodiment 1 of the present invention. FIG. **2**A is a plan view of optical connector **100**, and FIG. **2**B is a sectional view taken along line A-A illustrated in FIG. **2**A. Note that pressing member **135** and optical transmission member **120** are omitted in FIG. **1**, and optical transmission member **120** is illustrated with the solid line in FIGS. **2**A and **2**B.

Note that in the following description, the direction in which optical transmission members **120** are disposed in parallel (the direction in which projecting surfaces **151** of second surface **150** are arranged) is "first direction" or "X direction", the direction (height direction) orthogonal to the X direction in front view of second surface **150** is "second direction" or "Z direction", and the direction orthogonal to the X direction and the Z direction is "third direction" or "Y direction".

As illustrated in FIGS. **1**, **2**A and **2**B, optical connector **100** according to Embodiment 1 includes ferrule **110** and a plurality of optical transmission members **120**. Optical connector **100** according to the present embodiment can be used as an optical connector module together with a housing, a spring clamp structure part and the like.

The type of optical transmission member **120** is not limited. Examples of the type of optical transmission member **120** include optical fibers and optical waveguides. In the present embodiment, optical transmission member **120** is an optical fiber. In addition, the optical fiber may be of a single mode type, or a multiple mode type. Preferably, the end surface of optical transmission member **120** is tilted with respect to the plane orthogonal to the extending direction of optical transmission member **120**. In the present embodiment, the inclination angle to the plane is 5 degrees. The number of optical transmission members **120** is not limited as long as a plurality of optical transmission members **120** is provided. In the present embodiment, the number of optical transmission members **120** is 14. The end portion of optical transmission member **120** is held by ferrule **110**.

Ferrule **110** is a member with a substantially cuboid shape, and holds the front end portion of optical transmission member **120**. Ferrule **110** includes holding part **130**, first surface **140**, and second surface **150**. Ferrule **110** is formed with a material that is optically transparent to wavelengths used for optical communications. Examples of the material of ferrule **110** include polyetherimide (PEI) such as ULTEM (registered trademark) and transparent resins such as cyclic olefin resin. In addition, ferrule **110** may be manufactured by injection molding. Ferrule **110** may be used for the purpose of transmission, reception, or transmission and reception.

Holding part **130** holds the end portion of optical transmission member **120**. Holding part **130** includes first holding recess **131**, at least one through hole **132**, and a plurality of first grooves **133**. Note that in the present embodiment, holding part **130** includes second holding recess **134** and pressing member **135** in addition to the above-mentioned configurations.

First holding recess **131** includes first surface **140** and third surface **136***a* facing first surface **140**, and is open at the top surface of ferrule **110**. The plan shape of first holding recess **131** is not limited as long as the end portions of the plurality of optical transmission members **120** can be disposed at appropriate positions. In the present embodiment, the plan shape of first holding recess **131** is rectangular. First surface **140** is disposed at one inner surface in the third direction (the Y direction) of first holding recess **131**. Third surface **136***a* is disposed at the other inner surface in the third direction (the Y direction) of first holding recess **131**. First holding recess **131**, which is disposed between first surface **140** and third surface **136***a*, defines the extending direction of optical transmission member **120**. Third surface **136***a* is a part of wall **136** where through hole **132** opens. In the present embodiment, first groove **133** and groove part **137** are disposed in the bottom surface of first holding recess **131**.

Optical transmission member **120** is inserted to at least one through hole **132**. Through hole **132** guides optical transmission member **120** to first groove **133** and prevents positional displacement with respect to first groove **133**. Through hole **132** opens at first surface **140** and third surface **136***a* facing first surface **140** of first holding recess **131**, and is disposed along the extending direction of optical transmission member **120** (the extending direction of first groove **133**). That is, through hole **132** is disposed along the third direction (the Y direction). At least one through hole **132** is provided, and a plurality of through holes **132** may be provided. In the case where one through hole **132** is provided, through hole **132** may be a long hole, or may have a shape composed of a plurality of columnar through holes partially connected with each other. In the case where a plurality of through holes **132** is provided, the number of through holes **132** need only be greater than the number of optical transmission members **120** installed. In the present embodiment, the number of through holes **132** is the same as the number of optical transmission members **120**. Specifically, in the present embodiment, the number of through holes **132** is 14.

Through hole **132** may be disposed such that its central axis is parallel to the rear surface of ferrule **110**, or that it comes closer to the rear surface of ferrule **110** in the direction toward first surface **140**. In the present embodiment, through hole **132** is disposed parallel to the rear surface of ferrule **110**.

The size of through hole **132** is not limited as long as optical transmission member **120** can be inserted. Preferably, the minimum length of the opening of at least one through hole **132** is two or more times greater than the depth of first groove **133**. In addition, preferably, the minimum length of the opening of at least one through hole **132** is greater than the diameter of optical transmission member **120**. More specifically, preferably, the minimum length of the opening of at least one through hole **132** is 1.008 times or more greater than the diameter of optical transmission member **120**. The shape of the opening of through hole **132** openning at third surface **136***a* and the shape of the opening of through hole **132** openning at fourth surface **136***b* are not limited as long as optical transmission member **120** can be inserted. The shape of the opening of through hole **132** openning at third surface **136***a* and the shape of the opening of through hole **132** openning at fourth surface **136***b* may be a circular shape, an elliptical shape, or a polygonal shape. In the present embodiment, the shape of the opening of through hole **132** openning at third surface **136***a* and the shape of the opening of through hole **132** openning at fourth surface **136***b* are each a circular shape. In addition, preferably, the minimum length of at least one through hole **132** in the XZ cross-section is two or more times greater than the depth of first groove **133**. The shape of through hole **132** in the XZ cross-section is not limited as long as optical transmission member **120** can be inserted. The shape of through hole **132** in the XZ cross-section may be a circular shape, an elliptical shape, or a polygonal shape. In the present embodiment, the shape of through hole **132** in the XZ cross-section is a circular shape. In through hole **132**, the size from the opening of through hole **132** openning at third surface **136***a* to the opening of through hole **132** openning at fourth surface **136***b* may be the same or differ. More specifically, for example, it may be disposed such that the size decreases from the opening on the back side (second holding recess **134** side) of ferrule **110** toward the opening on the front surface side (first holding recess **131** side) of ferrule **110**. In addition, preferably, the plurality of (**14**) through holes **132** is disposed at respective positions corresponding to the plurality of first grooves **133**.

The plurality of first grooves **133** is disposed along the extending direction of optical transmission member **120** in the bottom surface of first holding recess **131**, and optical transmission member **120** inserted in through hole **132** is disposed thereto. First groove **133** may be disposed over the entire bottom surface of first holding recess **131**, or in a part of the bottom surface of first holding recess **131**. In the present embodiment, first groove **133** is disposed in a region of a part of the bottom surface of first holding recess **131** on first surface **140** side. The number of first grooves **133** need only be equal to or greater than the number of optical transmission members **120** installed. In the present embodiment, the number of first grooves **133** is the same as the number of optical transmission members **120**. Specifically, in the present embodiment, the number of first grooves **133** is 14. The cross-sectional shape of first groove **133** in the XZ cross-section is not limited. First groove **133** may be a V-shaped groove, or a U-shaped groove. In the present embodiment, first groove **133** is a V-shaped groove. Here, "V-groove" refers to a groove composed of two planes with a v shape in the cross-section perpendicular to the extending direction of the groove. The connecting part of two planes may include another plane between two planes, or the connecting part of adjacent first grooves **133** may be connected by a curved surface. The "U-groove" refers to a groove composed of one curved surface with an arc shape in the cross-section perpendicular to the extending direction of the groove. Preferably, the depth of first groove **133** is a depth with which the upper end portion of optical transmission member **120** is located above the upper end portion of first groove **133** (ridge) in the state where optical transmission member **120** is disposed at first groove **133**. In this manner, removal of optical transmission member **120** can be prevented by pressing optical transmission member **120** toward first groove **133** with pressing member **135**.

First groove **133** may be disposed parallel to the rear surface of ferrule **110**, or may be disposed such that it comes closer to the rear surface of ferrule **110** in the direction toward first surface **140**. In the present embodiment, first groove **133** is disposed parallel to the rear surface of ferrule **110**.

Here, a positional relationship between first groove **133** and through hole **132** is described. FIGS. **3**A to **5**C are schematic views illustrating a positional relationship between first groove **133** and through hole **132**. FIGS. **3**A to **3**C illustrate a case where first groove **133** is a V-groove and through hole **132** is a plurality of through holes, FIGS. **4**A to **4**C illustrate a case where first groove **133** is a V-groove and through hole **132** is one long hole, and FIGS. **5**A to **5**C illustrate a case where through hole **132** is a plurality of through holes and first groove **133** is a U-shaped groove. In FIGS. **3**A to **5**C, through hole **132** is indicated with solid line, and first groove **133** is indicated with broken line.

FIGS. **3**A, **4**A and **5**A illustrate a case where distance d is not provided between through hole **132** and first groove **133** as viewed along the extending direction of optical transmission member **120**, and FIGS. **3**B, **3**C, **4**B, **4**C **5**B and **5**C illustrate a case where distance d is provided between through hole **132** and first groove **133** as viewed along the extending direction of optical transmission member **120**.

As illustrated in FIGS. **3**A to **3**C and **5**A to **5**C, preferably, the position of the opening of through hole **132** on first holding recess **131** side and the position of first groove **133** in the first direction (the X direction) is disposed such that the center of the opening and the valley line of first groove **133** coincide with each other. Further, preferably, they are disposed such that optical transmission member **120** in through hole **132** and optical transmission member **120** on first groove **133** are located on the same straight line in the first direction (the X direction) (omitted in the drawing).

As illustrated in FIG. **3**A, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part of first groove **133** may be in contact with each other. That is, as viewed along the extending direction of optical transmission member **120**, no distance may be provided between through hole **132** and first groove **133**.

As illustrated in FIG. **3**B, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part of first groove **133** may be separated from each other. In the example illustrated in FIG. **3**B, as viewed along the extending direction of optical transmission member **120**, through hole **132** and first groove **133** do not overlap each other. In this case, preferably, the distance between through hole **132** and first groove **133** is 20 μm or smaller, more preferably 10 μm or smaller, still more preferably 5 μm or smaller as viewed along the extending direction of optical transmission member **120**. The distance between through hole **132** and first groove **133** means the shortest distance between through hole **132** and first groove **133** as viewed along the extending direction of optical transmission member **120**.

As illustrated in FIG. **3**C, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part of first groove **133** may overlap each other. In this case, preferably, the distance between through hole **132** and first groove **133** is 20 μm or smaller, more preferably 10 μm or smaller, still more preferably 5 μm or smaller as viewed along the extending direction of optical transmission member **120**. As illustrated in FIG. **3**C, in the case where through hole **132** overlaps first groove **133** as viewed along the extending direction of optical transmission member **120**, the inserted optical transmission member may hit the groove, and the ferrule may possibly be scraped. In view of this, in the case where through hole **132** and first groove **133** are not in contact with each other, it is preferable that through hole **132** do not overlap first groove **133** as illustrated in FIG. **3**B.

As described above, when the distance between through hole **132** and first groove **133** is 20 μm or smaller, optical transmission members **120** inserted from respective through holes **132** can be guided to corresponding first grooves **133**. Preferably, the distance between through hole **132** and first groove **133** is 0 mm, i.e., through hole **132** and first groove **133** are in contact with each other as viewed along the extending direction of optical transmission member **120**.

As illustrated in FIGS. **4**A to **4**C, in the case where through hole **132** is a single long hole, a part of through hole **132** and a part of first groove **133** may be or may not be in contact with each other as viewed along the extending direction of optical transmission member **120**. Also in this case, preferably, the distance between through hole **132** and first groove **133** is 20 μm or smaller, more preferably 10 μm or smaller, still more preferably 5 μm or smaller as viewed along the extending direction of optical transmission member **120**.

As illustrated in FIG. **5**A, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part of first groove **133** may be in contact with each other. Here, a state where first groove **133** and through hole **132** are in contact with each other means a state where the bottom portion of first groove **133** and the bottom portion of through hole **132** are in contact with each other. In this case, the opening of first groove **133** overlaps through hole **132**.

As illustrated in FIG. **5**B, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part (opening) of first groove **133** are in contact with each other.

As illustrated in FIG. **5**C, as viewed along the extending direction of optical transmission member **120**, a part of through hole **132** and a part of first groove **133** may overlap each other. In the example illustrated in FIG. **5**C, as viewed along the extending direction of optical transmission member **120**, through hole **132** and the bottom portion and the opening of first groove **133** overlap each other. In this case, preferably, the distance between through hole **132** and first groove **133** is 20 μm or smaller, more preferably 10 μm or smaller, still more preferably 5 μm or smaller as viewed along the extending direction of optical transmission member **120**.

In this manner, when the distance between through hole **132** and first groove **133** is 20 μm or smaller as viewed along the extending direction of optical transmission member **120**, optical transmission member **120** inserted to through hole **132** does not make contact with first groove **133**.

Note that in the present embodiment, groove part **137** with a depth greater than the depth of first groove **133** is disposed between first groove **133** and through hole **132**. Groove part **137** extends in the first direction (the X direction). In this case, preferably, the bottom portion of through hole **132** is located above the bottom portion of first groove **133** as illustrated in FIGS. **3**A, **4**A and **5**C.

Second holding recess **134** includes fourth surface **136***b*, and is disposed such that wall **136** is located between it and first holding recess **131**. Second holding recess **134** is open at the back surface and the top surface of ferrule **110**. The plan shape of second holding recess **134** is not limited as long as the plurality of optical transmission members **120** can be installed. In the present embodiment, the plan shape of second holding recess **134** is a rectangular shape. Second holding recess **134** defines the end portion in the extending direction of optical transmission member **120** on the back side of ferrule **110** than fourth surface **136***a*. Through hole **132** opens at fourth surface **136***b* on one side (front surface side of ferrule **110**) of second holding recess **134** in the third direction (the Y direction), and the other side thereof (the back side of ferrule **110**) is open.

Pressing member **135** presses optical transmission member **120** toward first groove **133**. In other words, pressing member **135** presses optical transmission member **120** toward a connector main body including first surface **140**, second surface **150**, first holding recess **131** and through hole **132**. Pressing member **135** is housed in first holding recess **131** where the end portion of optical transmission member **120** is disposed, and fixed with an adhesive and the like.

First surface **140** is disposed to face the end surfaces of the plurality of optical transmission members **120** held at holding part **130**. First surface **140** allows incidence of light emitted from the plurality of optical transmission members **120**. Note that first surface **140** may emit, toward the end surfaces of the plurality of optical transmission members **120**, light entered from second surface **150**. The shape of first surface **140** is not limited as long as the above-mentioned function can be ensured. First surface **140** may include a plurality of projecting surfaces or may be a flat surface. In the present embodiment, first surface **140** is a flat surface. First surface **140** is disposed at a part of the inner surface in the third direction (the Y direction) of first holding recess **131**.

The surface of first surface **140** that makes contact with the end surface of optical transmission member **120** may be tilted such that it comes closer to second surface **150** in the direction toward the rear surface of ferrule **110**, or may be perpendicular to the rear surface of ferrule **110**. In the present embodiment, the surface of first surface **140** that makes contact with the end surface of optical transmission member **120** is disposed such that it comes closer to second surface **150** in the direction toward the rear surface of ferrule **110**. Preferably, the inclination angle of first surface **140** is the same as the inclination angle of the end surface of optical transmission member **120**. Preferably, the inclination angle of first surface **140** with respect to the second direction (the Z direction) set as 0 degree is 3 to 8 degrees, more preferably 5 to 8 degrees, for example. In the present embodiment, the inclination angle of first surface **140** with respect to the second direction (the Z direction) set as 0 degree is 5 degrees.

Second surface **150** emits, to the outside, the light entered from first surface **140**. Note that second surface **150** may allow incidence of light from the outside. The shape of second surface **150** is not limited as long as the above-mentioned function can be ensured. Second surface **150** may include a plurality of projecting surfaces or may be a flat surface. In the present embodiment, second surface **150** includes a plurality of projecting surfaces (optical control surfaces) **151**. Projecting surfaces **151**, disposed in parallel in the first direction (the X direction), emit, toward another ferrule **110**, the light entered from first surface **140**, or allow incidence of light from the outside. Second surface **150** is disposed at the front surface of ferrule **110**. The plan shape of projecting surface **151** is not limited. The plan shape of projecting surface **151** may be a circular shape or a rectangular shape. In the present embodiment, the plan shape of projecting surface **151** is a circular shape. In addition, the number of projecting surfaces **151** is the same as the number of optical transmission members **120**. Specifically, in the present embodiment, the number of projecting surface **151** is 14.

Now, a method of attaching optical transmission member **120** to ferrule **110** is described. First, the plurality of optical transmission members **120** is inserted to through hole **132** from the opening of fourth surface **136***b* of ferrule **110**. Next, optical transmission members **120** inserted to through hole **132** are disposed on respective first grooves **133**, the end surfaces of the plurality of optical transmission members **120** is brought into contact with first surface **140**. In this state, adhesive is applied to the front end portion of optical transmission member **120** in contact with first surface **140**. Next, pressing member **135** is housed in first holding recess **131** in such a manner as to press optical transmission member **120** toward first groove **133**. Finally, adhesive is cured to fix optical transmission member **120** to ferrule **110**. In this manner, through hole **132** serves as a guide for placing optical transmission member **120**, and thus each optical transmission member **120** can be appropriately installed to each first groove **133**.

Effects

In the above-mentioned manner, according to the present embodiment, ferrule **110** includes through hole **132** and first groove **133**, and thus the plurality of optical transmission members **120** can be easily appropriately installed to ferrule

110. Thus, the front end of optical transmission member **120** can be accurately positioned and it can be can be easily assembled.

Embodiment 2

Next, optical connector **200** according to Embodiment 2 is described. Optical connector **200** according to the present embodiment is different from optical connector **100** according to Embodiment 1 only in ferrule **210**. The components similar to those of optical connector **100** according to Embodiment 1 are denoted with the same reference numerals and the description thereof is omitted.

Configuration of Optical Connector

FIG. **6** is a perspective view of optical connector **200** according to Embodiment 2 of the present invention. FIG. **7**A is a plan view of optical connector **200**, and FIG. **7**B is a sectional view taken along line A-A illustrated in FIG. **7**A. Note that pressing member **135** and optical transmission member **120** are omitted in FIG. **6**, and optical transmission member **120** is indicated with solid line in FIGS. **7**A and **7**B.

As illustrated in FIGS. **6**, **7**A and **7**B, optical connector **200** according to Embodiment 2 includes ferrule **210** and optical transmission member **120**.

Ferrule **210** includes holding part **230**, first surface **140**, second surface **150**, second holding recess **134**, pressing member **135**, and a plurality of second grooves **238**.

The plurality of second grooves **238** is disposed at second holding recess **134** along the extending direction of optical transmission member **120** in a manner corresponding to at least one through hole **132**. Second groove **238** guides optical transmission member **120** to first groove **133**. Second groove **238** may be disposed in the entire bottom surface of second holding recess **134**, or in a part of the bottom surface of second holding recess **134**. In the present embodiment, second groove **238** is disposed in the region of a part of the bottom surface of second holding recess **134** on fourth surface **136***b* side. In addition, in the present embodiment, second groove **238** is disposed in contact with wall **136** (fourth surface **136***b*). A plurality of second grooves **238** is provided, and the number of second grooves **238** need only be equal to or greater than the number of optical transmission members **120** installed. In the present embodiment, the number of second grooves **238** is the same as the number of first grooves **133** and the number of through holes **132**. Specifically, the number of second groove **238** is 14.

Preferably, the width of second groove **238** is greater than the minimum length of the opening of through hole **132**. The cross-sectional shape of second groove **238** in the XZ cross-section is not limited. Second groove **238** may be a V-shaped groove, or a U-shaped groove. In the present embodiment, second groove **238** is a U-shaped groove, and its cross-sectional shape is a semicircular shape. The depth of second groove **238** is not limited as long as optical transmission member **120** can be guided to through hole **132**. In the present embodiment, the depth of second groove **238** is the same as the radius of through hole **132**. In this manner, no step is formed at the boundary between the bottom portion of second groove **238** and through hole **132**, and thus optical transmission member **120** can be appropriately guided to through hole **132**.

Preferably, the position of the opening of through hole **132** on second holding recess **134** side and the position of second groove **238** in the second direction (the Z direction) are set such that the center of the opening and the bottom portion of second groove **238** coincide with each other. Preferably, they are disposed such that optical transmission member **120** in through hole **132** and optical transmission member **120** on second groove **238** are located on the same straight line in the second direction (the Z direction).

In the present embodiment, to attach optical transmission member **120** to ferrule **210**, first, each optical transmission member **120** is guided to each through hole **132** with second groove **238** as a guide. Next, the plurality of optical transmission members **120** is inserted to through hole **132** from the opening of fourth surface **136***b* of ferrule **210**. Next, optical transmission members **120** inserted to through hole **132** are disposed on respective first grooves **133**, and the end surfaces of the plurality of optical transmission members **120** are brought into contact with first surface **140**. In this state, adhesive is applied to the front end portion of optical transmission member **120** in contact with first surface **140**. Next, pressing member **135** is housed in first holding recess **131** in such a manner as to press optical transmission member **120** against first groove **133**. Finally, the adhesive is cured to fix optical transmission member **120** to ferrule **110**. In this manner, through hole **132** functions as a guide for placing optical transmission member **120**, and thus each optical transmission member **120** can be appropriately installed to each first groove **133**.

Effects

In the above-mentioned manner, according to the present embodiment, second groove **238** is further provided, and thus it can be further easily assembled in comparison with optical connector **100** according to Embodiment 1.

Reference Example

Next, optical connector **500** according to a reference example is described. Optical connector **500** according to the reference example is the same as optical connector **200** according to Embodiment 2 except that wall **136** is not provided. Therefore, the components similar to optical connector **200** according to Embodiment 2 are denoted with the same reference numerals, and the description thereof is omitted.

Configuration of Optical Connector

FIG. **8** is a perspective view of optical connector **500** according to the reference example. FIG. **9**A is a plan view of optical connector **100**, and FIG. **9**B is a sectional view taken along line A-A illustrated in FIG. **9**A. Note that pressing member **135** and optical transmission member **120** are omitted in FIG. **8**, and optical transmission member **120** is indicated with solid line in FIGS. **9**A and **9**B.

As illustrated in FIGS. **8**, **9**A and **9**B, optical connector **500** according to the reference example includes ferrule **510** and optical transmission member **120**.

Ferrule **510** includes holding part **530**, first surface **140**, second surface **150**, second holding recess **134**, pressing member **135**, and the plurality of second grooves **238**.

Effects

According to the reference example, since second groove **238** is provided, each optical transmission member **120** can be easily disposed to each first groove **133**. In addition, since no through hole **132** is provided, the assembly accuracy is lower than in Embodiments 1 and 2.

Industrial Applicability

The ferrule, optical connector and optical connector module according to the present invention are suitable for optical communications using optical transmission members.

Industrial Applicability

100, **200**, **500** Optical connector
110, **210**, **510** Ferrule
120 Optical transmission member
130, **230**, **530** Holding part
131 First holding recess
132 Through hole
133 First groove
134 Second holding recess
135 Pressing member
136 Wall
136*a* Third surface
136*b* Fourth surface
137 Groove part
238 Second groove
140 First surface
150 Second surface

What is claimed is:

1. A ferrule configured to hold a plurality of optical transmission members, the ferrule comprising:
- a holding part configured to hold one end portion of the plurality of optical transmission members;
- a first surface configured to allow, to enter the ferrule, light emitted from the plurality of optical transmission members held by the holding part, or configured to emit, toward the plurality of optical transmission members, light advanced inside the ferrule; and
- a second surface configured to emit, to outside of the ferrule, the light advanced inside the ferrule, or configured to allow, to enter the ferrule, light from the outside,

wherein the holding part includes:
- a first holding recess including the first surface and a third surface facing the first surface,
- at least one through hole that is open at the third surface and a fourth surface, the fourth surface being located on a side on which the plurality of optical transmission members is inserted, and
- a plurality of first grooves disposed at the first holding recess along an extending direction of the plurality of optical transmission members, the plurality of optical transmission members inserted to the at least one through hole is respectively disposed at the plurality of first grooves,
- a second holding recess including the fourth surface, and
- a plurality of second grooves disposed at the second holding recess along the extending direction of the plurality of optical transmission members in a manner corresponding to the plurality of first grooves, the plurality of second grooves contacting with the fourth surface.

2. The ferrule according to claim **1**, wherein the second surface includes a plurality of optical control surfaces configured to emit, to the outside of the ferrule, light emitted from the plurality of optical transmission members and advanced inside the ferrule, or configured to allow, to enter the ferrule, the light from the outside.

3. The ferrule according to claim **1**, wherein a minimum length of an opening of the at least one through hole is two or more times greater than a depth of the plurality of first groove.

4. The ferrule according to claim **1**, wherein a minimum length of an opening of the at least one through hole is greater than a diameter of the plurality of optical transmission members.

5. The ferrule according to claim **1**, wherein:
- the at least one through hole includes a plurality of through holes; and
- the plurality of through holes is respectively disposed at positions corresponding to the plurality of first grooves.

6. The ferrule according to claim **5**, wherein as viewed along the extending direction of the plurality of optical transmission members, a distance between each of the plurality of through holes and each of the plurality of first grooves is 20 μm or smaller.

7. The ferrule according to claim **1**, further comprising a pressing member configured to press, toward the plurality of first grooves, the plurality of optical transmission members disposed on the plurality of first grooves.

8. An optical connector, comprising:
- the ferrule according to claim **1**; and
- a plurality of optical transmission members held by the ferrule.

9. An optical connector module, comprising the optical connector according to claim **8**.

* * * * *